

Aim: Implement BFS and DFS in a graph.

1.Depth First Search (DFS).

CODE:

```
#include <bits/stdc++.h>
using namespace std;
void dfsRec(vector<vector<int> > &adj, vector<bool> &visited, int s, vector<int> &res){
    visited[s] = true;
    res.push_back(s);
    for(int j = 0; j < adj[s].size(); j++){
        int i = adj[s][j];
        if (visited[i] == false)
            dfsRec(adj, visited, i, res);
    }
}
vector<int> DFS(vector<vector<int> > &adj){
    vector<bool> visited(adj.size(), false);
    vector<int> res;
    dfsRec(adj, visited, 0, res);
    return res;
}
void addEdge(vector<vector<int> > &adj, int s, int t){
    adj[s].push_back(t);
    adj[t].push_back(s);
}

int main(){
    int V, E;
    cout << "Enter number of vertices: ";
    cin >> V;
    cout << "Enter number of edges: ";
    cin >> E;
    vector<vector<int> > adj(V);
    cout << "Enter " << E << " edges (u v):" << endl;
    for(int i = 0; i < E; i++){
        int u, v;
        cin >> u >> v;
        addEdge(adj, u, v);
    }
    vector<int> res = DFS(adj);
    cout << "DFS starting from node 0: ";
    for(int j = 0; j < res.size(); j++){
        cout << res[j] << " ";
    }
}
```

OUTPUT:

```

C:\Users\ROHIT\OneDrive\De  X + v
Enter number of vertices: 5
Enter number of edges: 5
Enter 5 edges (u v):
1 2
1 0
2 0
2 3
2 4
DFS starting from node 0: 0 1 2 3 4
-----
Process exited after 6.094 seconds with return value 0
Press any key to continue . . . |

```

2. Breadth First Search (BFS).

CODE:

```

#include <bits/stdc++.h>
using namespace std;

```

```

vector<int> bfs(vector<vector<int> > &adj){
    int V = adj.size();
    int s = 0;
    vector<int> res;
    queue<int> q;
    vector<bool> visited(V, false);
    visited[s] = true;
    q.push(s);
    while(!q.empty()){
        int curr = q.front();
        q.pop();
        res.push_back(curr);
        for(int j = 0; j < adj[curr].size(); j++){
            int x = adj[curr][j];
            if (!visited[x]){
                visited[x] = true;
                q.push(x);
            }
        }
    }
    return res;
}

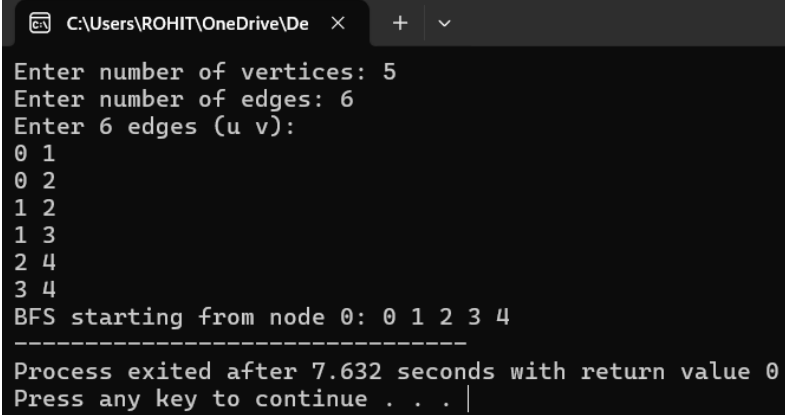
```

```

int main(){
    int V, E;
    cout << "Enter number of vertices: ";
    cin >> V;
    cout << "Enter number of edges: ";
    cin >> E;
    vector<vector<int> > adj(V);
    cout << "Enter " << E << " edges (u v):" << endl;
    for(int i = 0; i < E; i++){

```

```
    int u, v;  
    cin >> u >> v;  
    adj[u].push_back(v);  
    adj[v].push_back(u);  
}  
vector<int> ans = bfs(adj);  
cout << "BFS starting from node 0: ";  
for(int i = 0; i < ans.size(); i++){  
    cout << ans[i] << " ";  
}  
return 0;  
}
```

OUTPUT:

```
C:\Users\ROHIT\OneDrive\De  X  +  v  
Enter number of vertices: 5  
Enter number of edges: 6  
Enter 6 edges (u v):  
0 1  
0 2  
1 2  
1 3  
2 4  
3 4  
BFS starting from node 0: 0 1 2 3 4  
-----  
Process exited after 7.632 seconds with return value 0  
Press any key to continue . . . |
```